

DMLT - DIPLOMA IN MEDICAL LABORATORY TECHNOLOGY

SYLLABUS

SCHEME OF EXAMINATION

Subject code	Title of the Course	Hours	Passing Minimum
Semester I			
Subject 1	Anatomy and Physiology	3 Hours	40/100
Subject 2	Blood Transfusion and Blood Banking	3 Hours	40/100
Subject 3	Clinical Haematology	3 Hours	40/100
Subject 4	Communicative English	3 Hours	40/100
Subject 5	Practical I-Lab in Analytical Biochemistry and Haematological Techniques	3 Hours	40/100
Semester II			
Subject 1	Clinical Microbiology	3 Hours	40/100
	Clinical Biochemistry	3 Hours	40/100
Subject 2	Life Skill	3 Hours	40/100
Subject 3	Industrial Visit and Internship	3 Hours	40/100
Subject 4	Practical II-Lab in Analytical Biochemistry II	3 Hours	40/100
Semester III			
Subject 1	Principles of Laboratory Management & Medical Ethics	3 Hours	40/100
Subject 2	Histopathology	3 Hours	40/100
Subject 3	Clinical Pathology	3 Hours	40/100
Subject 4	Standard Operating Procedures and Safety Practices	3 Hours	40/100
Subject 5	Practical III-Lab in Microbial Techniques	3 Hours	40/100
Semester IV			
Subject 1	Modern Instrumentation in Diagnostic Lab	3 Hours	40/100
Subject 2	Bio-Medical Waste Management	3 Hours	40/100
Subject 3	Clinical Endocrinology & Toxicology	3 Hours	40/100
Subject 4	Practical IV-Lab in Endocrinology, Cancer Diagnosis & Molecular Diagnosis	3 Hours	40/100
Subject 5	Project Work	3 Hours	40/100

Eligibility for admission: Pass in 10th std examination conducted by the Govt. of Tamil Nadu Board of Secondary Education, Government of Tamil Nadu or any other equivalent examination.

Examination: Passing Minimum for each Course is 40%. Classification will be done on the basis of percentage marks of the total marks obtained in all the Courses and as given below:

- 40 % but less than 50 % - Third class
- 50 % but less than 60 % - Second class
- 60 % and above - First class

Theory Paper

Internal Marks-25

External Marks-75

SYLLABUS

Semester I

- Subject 1 : Anatomy and Physiology
- Subject 2 : Blood Transfusion and Blood Banking
- Subject 3 : Clinical Haematology
- Subject 4 : Communicative English
- Subject 5 : Practical I-Lab in Analytical Biochemistry and Haematological Techniques

Semester II

- Subject 1 : Clinical Microbiology
- Subject 2 : Clinical Biochemistry
- Subject 3 : Life Skill
- Subject 4 : Industrial Visit and Internship
- Subject 5 : Practical II-Lab in Analytical Biochemistry II

Semester III

- Subject 1 : Principles of Laboratory Management & Medical Ethics
- Subject 2 : Histopathology
- Subject 3 : Clinical Pathology
- Subject 4 : Standard Operating Procedures and Safety Practices
- Subject 5 : Practical III-Lab in Microbial Techniques

Semester IV

- Subject 1 : Modern Instrumentation in Diagnostic Lab
- Subject 2 : Bio-Medical Waste Management
- Subject 3 : Clinical Endocrinology & Toxicology
- Subject 4 : Practical IV-Lab in Endocrinology, Cancer Diagnosis & Molecular Diagnosis
- Subject 5 : Project Work

SEMESTER I
Subject I
ANATOMY AND PHYSIOLOGY

UNIT I **18 Hrs**
GENERAL ANATOMY

Typical animal cell (Structure & Function) –primary tissues (Classification & function)
Skeletal System-Digestive System- Functions of stomach-composition of gastric juice-
Pancreatic Juice-Bile and Digestion of food by different Enzymes-Absorption and
Defecation.

UNIT II **18 Hrs**
RESPIRATORY AND DIGESTIVE SYSTEMS

Nose-Trachea-Bronchi Lungs and Pleura-Digestive System-Alimentary canal (different
parts)-Liver-Gall Bladder-Pancreases.

UNIT III **18 Hrs**
UROGENITAL SYSTEM

Different parts of urinary system-Different parts of Male & Female genital -System
(Internal & External Genitalia) Special Senses& General Sensibilities- Eye & Vision-
Ears-Hearing & Equilibrium-Excretory System-Functions of Kidney-Nephron and
Function of Skin.

UNIT IV **18 Hrs**
NERVOUS & REPRODUCTIVE SYSTEMS

Brain & Spinal Cord - Autonomic Nervous System-Head & Neck-Thorax- Abd. &
Pelvis- Surface Anatomy-Reproductive System-Name of primary and accessory organs
in male and female-Name of secondary sexual characters in male and female-
Function of ovary-formation of ova, menstrual cycle- Functions of Testes- Fertilization
Vasectomy and tubectomy.

UNIT V **18 Hrs**
CIRCULATORY SYSTEM

Heart-Blood Vessels-Lymphatic –Spleen & Thymus. Blood- Groups-Composition and
general function of blood- Description of blood cells - normal counts & function-
Anticoagulants-Cerebrospinal Fluid-Formation-Composition & function of lymph-
Endocrine-glands- Names of the endocrine gland and the hormone secreted by them-
Major actions of such Hormones.

Subject II

BLOOD TRANSFUSION AND BLOOD BANKING

Objective:

- To understand the basics of blood transfusion and blood banking

Learning Outcome:

- Will understand the blood grouping tests
- Will understand the blood donor selection criteria
- Will know the blood collection and processing techniques
- Will be able to do compatibility tests, antibody screening and identification
- Will understand the blood components separation storage and transportation procedures.
- Will know the various agencies in blood donor and blood banking.

UNIT I

18 Hrs

Biochemistry and Physiology of Blood

Definition, Functions, Composition: Erythrocyte leucocyte and thrombocyte, Blood Haemoglobin, Methods and procedure for Haemoglobin test, ESR, PCV, Buffy coat, Total RBC Count, total WBC count, Differential WBC count, Total Eosinophil count, Total Platelet count.

UNIT II

18 Hrs

Quantitative and Qualitative Disorders of blood cells

Detailed study about Blood Coagulation factors, Bleeding Time, Clotting time, Prothrombin time. Anaemia: Definition, Types Of anaemia. Leukaemia Definition, Classification, Detailed Study with Lab finding for Myeloid Leukaemia (Acute & Chronic). Lymphatic Leukaemia (Acute & Chronic) Special topic: Abnormal forms of RBC, Haemophilia

UNIT III

18 Hrs

Blood Grouping and Blood Transfusion

Blood Grouping: Principle, Reagents, Methods and Procedure of Red Blood Cell and Serum Grouping with interpretation, Rh Typing, Importance Antigen, Antibody, Agglutination, Antigen - Antibody in different Blood Group, Sub Group of 'A' and 'AB', Bombay 'O' Blood Group, Preparation of RBC suspension. Blood transfusion: Types and indications of various blood transfusion, Universal donor / recipient.

UNIT IV

18 Hrs

Blood Collection and Compatibility Test

Criteria for selection, screening procedures, risks and management of donor complications, Blood collection procedures: vein puncture and finger prick, Anticoagulants used in Blood Bank, Pilot blood containers, Storage of donor blood, Basic procedures and Techniques for compatibility testing

UNIT V

18 Hrs

Blood transfusion hazards and their management

Nature, Types, and Investigation: Reaction investigation procedure with interpretation. Management of transfusion reactions Antiglobulin (Coomb's) Test, Haemolytic Disease of Newborn (HDN), Transmittable diseases Hepatitis, HIV Syphilis, Malaria - Detection and outline of their management.

Subject III

CLINICAL HAEMATOLOGY

UNIT I

18 Hrs

Collection of Blood Samples ABO Blood Grouping Procedure: Slide or Tile Method- Tube Method-Microplate Method-Micro-Typing System (Diamed/Bioview)-Automated or Semi-Automatic Instrumentation-Obtaining peripheral Blood Smear Staining of Blood Smear Obtaining Cell Counts – RBC-WBC-Platelets both manual and automated Absolute Eosinophils Count.

UNIT II

18 Hrs

Estimation of Haemoglobin Packed Cell Volume-Erythrocyte Indices Reticulocyte Count Differential Count Bleeding Time Clotting Time

UNIT III

18 Hrs

Glucose Determination: Oxidase Method Of Glucose Determination - The Colormetric Method--Ortho- Toluidine - The Glucose Tolerance Test (GTT) - Estimation of Serum Creatinine Biuret Method Bromocresol Green Method Modified Reitman & Frankel Method King & King Method.

UNIT IV

18 Hrs

Jaundice - Biochemical tests - Unconjugated Hyperbilirubinaemia (Retention Jaundice -Haemolytic (Pre-hepatic Jaundice) - Non haemolytic - Conjugated Hyperbilirubinaemia (Regurgitation Jaundice) Lipid profile - Total lipids - Phospholipids Sackett's Method Estimation of Serum HDL cholesterol Method of Fiske and Subbarow Caraway's Method of Estimation - Hyperuricaemia - Hypouricaemia

UNIT V

18 Hrs

Bile Salts: Hay's Test - For Bile Pigments - Fouchet's Test Collection- Appearance- Analysis of Cerebrospinal fluid- Synovial Fluid- Pleural Fluid- Pericardial Fluid- Peritoneal Fluid- Seminal Fluids- Needle aspiration Cytology- Discharge from any site.

Subject V
**PRACTICAL I-LAB IN ANALYTICAL BIOCHEMISTRY &
HEMATOLOGICAL TECHNIQUES**

Motion - Ova, Cyst, Trophozoite by wet smear preparation using normal saline and Lugol's iodine solution, Motion occult blood, microfilaria and Malarial Parasites - Identification in Stained blood smear, Special topic: ECG

SPECIAL TOPIC:

Haematuria, Pregnancy Test.

Blood group and Rh(D) Factor test by open slide method and test tube method, compatibility test (both major and minor) by saline technique for all available donors and patients.

Blood Glucose(GOD/POD) Blood/urine urea(Di acetyl Manoxime method), serum/urine creatinine (Alkaline Picrate Method), serum total cholesterol (Enzymatic Method), serum Total proteins (Bicrt Method), serum Albumin (Bromo cresol Greendye Method), serum Acid/Alkaline Phosphatase, serum Bilirubin, Glucose tolerance test.

Urine Specific gravity, Urine Albumin (Heat Coagulation Method), Urine Glucose (Benidict's Method), Urine Acetone (Nitroprusside Method), Urine Bile Salt (Hays Method), Urine Bile Pigments (Fouchest's Method), Microscopic Examination, Parasites, Pregnancy test (Latex Method), Urine reaction, Occult blood in Urine, Urobilinogen test

Subject IV
COMMUNICATIVE ENGLISH

1. **Basic Grammar:**
 - a. Review of grammar
 - b. Remedial study of grammar
 - c. Simple sentence
 - d. Word passive voice etc.
2. **Bubbling Vocabulary:**
 - a. Synonyms
 - b. Antonyms
 - c. One – word Institution
3. **Reading and Understanding English**
 - a. Comprehension passage
 - b. Précis – writing
 - c. Developing a story from hints.
4. **Writing English**
 - a. Writing Business letters.
 - b. Paragraph writing
 - c. Essay writing
 - d. Dialogue writing
5. **Speaking English**
 - a. Expressions used under different circumstances
 - b. Phonetics

SEMESTER II
Subject VI
CLINICAL MICROBIOLOGY

UNIT I **18 Hrs**

Microscopy

Parts and principles of simple microscope- compound microscope-phase contrast microscope.

UNIT II **18 Hrs**

Systemic Bacteriology

Definition-Classification-Staphylococcus-Streptococcus-Micrococci-Pneumococcus-Neisseria-Corynebacteria-Bacillus-Clostridium-Enterobacteriaceae-Klebsiella-Escherichia coli-Proteus-Salmonella-Shigella-Pseudomonas-Spirochetes.

UNIT III **18 Hrs**

Mycology

Classification of Fungus- Laboratory Diagnosis - Collection and transport of Specimen- Direct Microscopy-Classification of pathogenic Fungi:- Superficial Mycoses- Subcutaneous Mycoses-Systemic Mycoses-Opportunistic Mycoses.

UNIT IV **18 Hrs**

Virology

General properties of virus-Laboratory diagnosis of viral infections: Hepatitis virus-Human Immunodeficiency Virus-Polio Virus-Rabies Virus.

UNIT V **18 Hrs**

Parasitology

General Parasitology-Classification of parasites: Protozoa-Technical terms – parasite-Medical Parasitology-Host-Vector-Pathogen-Commensal-Ova-Cyst-Trophozoite-Cestode-Trematode-Nematode parasites- life cycle Pathogenicity- Lab Diagnosis and Morphology with Diagram of Entamoeba histolytica- Entamoeba coli-Giardia lamblia-Trichomonas vaginalis-Leishmania species- Malarial Parasites (Plasmodium Species)-Tapeworms-Round Worm-Hookworms-Microfilarial worms.

Subject VII CLINICAL BIOCHEMISTRY

UNIT I **18 Hrs**

INSTRUMENT

Detailed study about Photoelectric colorimeter- pH meter- Centrifuge- Analytical balance-Flame photometer- Analytical Balance-Use and Maintenance-cleaning of new and used glassware-Pipettes and test tubes.

UNIT II **18 Hrs**

ANALYTICAL PREPARATION OF SOLUTION REAGENTS

Definitions, types-Solute-Solvent-pH-Buffer-preparation of Phosphate Buffer Saline (PBS); Use and storage of buffer solution-Concentration of Solutions: Molarity-Normality-ppm-Dilution methods of Solution-Storage of Chemicals and Reagents- Safe Use Flammable Chemicals-Corrosive Chemicals- Toxic- Harmful and Irritating Chemicals- Oxidizing Chemicals- Explosive Chemicals-Carcinogens-Indicator-Oxidation-Reduction.

UNIT III **18 Hrs**

BIO-CHEMICAL TEST PROFILE

Liver Function tests-Renal Function Tests-Heart Function Tests-Pancreatic Function Tests-Diabetes-Jaundice-Lipids-Proteins.

UNIT IV **18 Hrs**

ROUTINE BIO-CHEMICAL TESTS

Blood Glucose-Blood / Urine Urea-Serum Bilirubin-Serum / Urine Creatinine-Glucose Tolerance Test-Serum total Cholesterol and high Density Lipoproteins-Serum total Proteins / Albumin / Globulin-SGOT-SGPT-Serum Alkaline / Acid Phosphatase- Serum Uric acid-Blood Urea Nitrogen Serum Calcium-Serum Amylase. Proteinuria Glucose-Ketone Bodies, Bile Pigments, Urobilinogen, Urobilin-Porphyrins- Haematuria-Iron-Calcium-Iodine-Flourine in Urine.

UNIT V **18 Hrs**

ELECTROLYTE TEST

Electrolytes with test procedure for photoelectric colorimetric method- (Na, K, Cl)- Quality control.

Subject VIII

LIFE SKILL

I Life Coping or adjustment

- (a) External and internal influence in one's life
- (b) Process of coping or adjustment
- (c) Coping with physical change and sexuality
- (d) Coping with stress, shyness, fear, anger far live and criticism.

II Attitude

- (a) Attitude
- (b) Self acceptance, self – esteem and self actualization
- (c) Positive thinking

III Problem Solving

- (a) Goal Setting
- (b) Decision Making
- (c) Time Management and stress Management.

IV Computers

- (a) Introduction to Computers
- (b) M.S.Office
- (c) Power Point

V Internet

- (a) Introduction to internet
- (b) E – mail
- (c) Browsing

Subject IX INDUSTRIAL VISIT

Students shall be taken for an visit to the an NABL Accreditation Diagnostic Laboratory and observe to various labs /Department wherein they can also undergo practical training. Know the quality policy and procedures practiced in the labs ad method of sample collection, methods of labeling , methods of analysis and reporting procedures. Student's has to submit an report to the teacher on the observation of visit.. The faculty shall submit the assessment records of each student .Marks will be awarded out of 100.

Subject X PRACTICAL II-LAB IN ANALYTICAL BIOCHEMISTRY II

SPECIAL TOPIC:

Haematuria, Pregnancy Test.

Blood group and Rh(D) Factor test by open slide method and test tube method, compatibility test (both major and minor) by saline technique for all available donors and patients.

Blood Glucose(GOD/POD) Blood/urine urea(Di acetyl Manoxime method), serum/urine creatinine (Alkaline Picrate Method), serum total cholesterol (Enzymatic Method), serum Total proteins (Bicrt Method), serum Albumin (Bromo cresol Greendye Method), serum Acid/Alkaline Phosphatase, serum Bilirubin, Glucose toterance test.

Urine Specific gravity, Urine Albumin (Heat Coagulation Method), Urine Glucose (Benidict's Method), Urine Acetone (Nitroprusside Method), Urine Bile Salt (Hays Method), Urine Bile Pigments (Fouchest's Method), Microscopic Examination, Parasites, Pregnancy test (Latex Method), Urine reaction, Occult blood in Urine, Urobilinogen test

SEMESTER III
Subject XI
PRINCIPLES OF LABORATORY MANAGEMENT & MEDICAL ETHICS

Objectives

- The students will be made aware of the basic ethics, good lab practices including awareness/ safety in a clinical lab.
- Students would be competent enough to understand sample accountability, quality management system, biomedical waste management, calibration and validation of clinical laboratory instruments, Laboratory Information system (LIS), Hospital Information system (HIS) and financial management.

UNIT I

18 Hrs

Ethical Principles and standards for a clinical laboratory professional duty to the patient-duty to colleagues and other professionals-Good Laboratory Practice (GLP)-Introduction to Basics of GLP and Accreditation-Aims of GLP and Accreditation-Advantages of Accreditation-Brief knowledge about National and International Agencies for clinical laboratory accreditation.

UNIT II

18 Hrs

Awareness/Safety in a clinical laboratory-General safety precautions-HIV: pre- and post-exposure guidelines-Hepatitis B & C: pre- and post-exposure guidelines-Drug Resistant Tuberculosis Patient management for clinical samples collection-transportation and preservation-Sample accountability-Purpose of accountability-Methods of accountability.

UNIT III

18 Hrs

Sample analysis: Introduction-factors affecting sample analysis-reporting results, basic format of a test report-reported reference range-clinical alerts-abnormal results-results from referral laboratories-release of examination results-alteration in reports.

UNIT IV

18 Hrs

Quality Management system: Introduction-Quality assurance-Quality control system, Internal and External quality control-quality control chart Biomedical Introduction and importance of calibration and Validation of Clinical Laboratory instrument Ethics in Medical laboratory Practice-Ethics in relation to Pre-Examination procedures-Examination procedures-reporting of results-preserving medical records Procurement of equipment and Inventory Control.

UNIT V

18 Hrs

Audit in a Medical Laboratory-Introduction and Importance-NABL & CAP-Responsibility-Planning-Horizontal-Vertical and Test audit-Frequency of audit-Documentation.

Subject XII HISTOPATHOLOGY

UNIT I **18 Hrs** **FIXATION**

Principles of Histology and its application-Principles of tissue fixation-Preparation of different fixatives: formaldehyde-Paraformaldehyde-methanol-acetone-decalcification-Detection of end point.

UNIT II **18 Hrs** **PROCESSING OF TISSUES**

Neutralization and processing of tissues (dehydration and rehydration)- Nuscum Techniques-manual tissue processing-automated tissue processing-paraffin embedding and different techniques for embedding-Preparation of paraffin blocks.

UNIT III **18 Hrs** **SECTIONING**

Microtome and its parts-Handling and care of microtome-sharpening and selection of razors and section cutting-Preparation of 1 micron thin section-frozen section/cryostat.

UNIT IV **18 Hrs** **STAINING**

Theory of staining -Common & special stains – Preparation of common stains. H & E-congored-methyle violet- Leishman stain- Giesma- Papiacolau-VG-PAS-PASM-Papanicolaous staining technique/MCC staining-staining techniques.

UNIT V **18 Hrs** **MOUNTING & OBSERVATION**

Different mounting solutions and protocols-Mounting of museum specimens-Microscoping observation-record keeping-indexing of slides-Histological observation of different tissues: skin-lungs-liver-urinogenetal and gastrointestinal tissues.

Subject XIII CLINICAL PATHOLOGY

UNIT I **18 Hrs**

SPECIMEN COLLECTION

Reception of patients-Phlebotomy and aftercare of patients-collection of different clinical materials-Blood-PUS-stool-sputum-semen-CSF-storage and Transportation of different clinical materials to different laboratories.

UNIT II **18 Hrs**

EXAMINATION OF URINE & STOOL

Parts of urinary system-importance of urine exam-Normal and abnormal contents of urine sample-Types of urine sample-urine collection and Preservatives.

Stool examination- indication-collection-container of Stool sample-transport-preservation of stool samples-physical examination and its significance-chemical examination and its significance-microscopic examination and its significance.

UNIT III **18 Hrs**

EXAMINATION OF SPUTUM

Sputum collection-container of Sputum sample-transport-storage of Sputum-examination-physical-chemical and microbial examinations and its significance.

UNIT IV **18 Hrs**

SEMEN ANALYSIS

Semen collection, container of semen sample-transport-storage of semen-examination- physical-chemical and microbial examinations and its significance.

UNIT V **18 Hrs**

EXAMINATION OF Cerebrospinal Fluid (CSF)

Collection of CSF-container of body fluids-transport of sample preservation of CSF-Examination of CSF and other body Fluids- fluid analysis-examination- physical-chemical and microbial examinations and its significance.

Subject XIV
STANDARD OPERATING PROCEDURES AND
SAFETY PRACTICES

UNIT I **18 Hrs**

SAFE HANDLING OF CHEMICALS AND EQUIPMENT

Operation and maintenance of laboratory equipments-Handling & cleaning of glassware (test tubes-slides petridishes pipettes-beakers, Rashes-funnels, syringes etc)-Collection & transport of clinical specimens-Receipts-Labeling-recording and dispatching clinical specimens-Keeping records after final computerization-Conversant with S.I. unit system for reporting-Conversant with Fundamental Chemistry.

UNIT II **18 Hrs**

SANITATION AND HYGIENE

Demonstrate knowledge of Good Laboratory Practices (GLPs)-Good Manufacturing Practices (GMPs) and Fire Safety-Statistical quality control-Using emergency equipment and Safety planning.

UNIT III **18 Hrs**

FIRST AID AND BASIC LIFE SUPPORT

Objectives of first aid-wounds and bleeding-dressing and bandages-pressure and splints-supports etc-shock-insensibility-asphyxia-convulsion-resuscitation-uses of suction apparatus-drug reactions-prophylactic measures-administration of oxygen, electric shock-burns-scalds -hemorrhage-pressure points-compression band. Fractures splints bandaging, dressing foreign bodies-poisons-Introduction to BLS-indication for BLS and the process of BLS-recovery position

UNIT IV **18 Hrs**

CHEMICAL SPILLS

Toxic/hazardous chemical spills- Cleanup procedures for chemical spills-hazardous chemical contact with the skin- safety procedures- maintenance of Emergency eye wash/safety showers - usage of Chemical Splash Goggles- Mouth pipetting or siphoning is forbidden- hazardous waste disposal.

UNIT V **18 Hrs**

DISPOAL OF LABORATORY WASTE

Introduction to laboratory Waste-Types of laboratory Waste-Collection of laboratory Waste-identification of all types of Treatment and Safe Disposal of laboratory Waste-non infectious waste-infected sharp waste-infected non sharp waste disposal-incineration process.

Subject XV
PRACTICAL III-LAB IN MICROBIAL TECHNIQUES

List of Exercises

- Preparation and sterilization of Nutrient Media,
- Inoculation Techniques, Pure culture, Sub culture,
- To perform Gram staining
- To perform Acid fast staining (Zeihl Neelsen staining)
- To perform Indian ink staining
- To perform Hanging drop method
- Demonstration of capsule
- Staining of bacterial spores
- Demonstration of Autoclave and sterilization of media
- Demonstration of Laminar air flow and media preparation
- Preparation of culture plates
- Demonstration of Centrifuge.
- Demonstration of hot air Oven and sterilization of glassware's
- Demonstration of Incubator and preservation of cultures
- Preparation of media
- Antibiotic sensitivity test.
- Microscopic examination of urine
- Examination of urine
- Examination of sputum
- To perform HIV Tridot test.

SEMESTER IV
Subject XVI
MODERN INSTRUMENTATION IN DIAGNOSTIC LAB

Course Objectives

- To gain knowledge on the modern instruments and techniques used in diagnostic lab.
- Will Understand the principles and application of spectrophotometric and Calorimetric equipments.
- Will Know the types of auto analyzers.
- Will Understand the Principles and applications of electrophoresis and Chromatography.
- Will Know the radio assay and Immuno assay
- Will understand the modern molecular techniques in disease diagnosis.

UNIT I

18 Hrs

Spectro Photometry

Principles and application of Spectro Photometer-Colorimeters-flame Photometers-fluoro cytometry.

UNIT II

18 Hrs

Auto analyzers

Principles and application of auto analyzer- Semi auto analyzer – Batch analyzer- Random assays- Auto analyzer.

UNIT III

18 Hrs

Electrophoresis and Chromatography

Chromatography and its applications in diagnosis- (Paper Chromatography- Thin layer Chromatography-HPLC-Gas liquid Chromatography-Ion exchange Chromatography)- Electrophoresis Basic Principles and types- Electrophoresis of Proteins and nucleic acids- Hemoglobin-Immunoglobulin's Iso enzymes- Application of electrophoresis in Clinical Diagnosis.

UNIT IV

18 Hrs

Immuno assay and Radio assay

Immuno assay – ELISA, RIA, FIA, FACS – Radio Isotopes, Radio activity-Instruments for Radio activity measurement application of radio Isotopes in Clinical biochemistry, blood volume-red cell volume-plasma volume-red cell lifespan-platelet lifespan.

Unit V

18 Hrs

Molecular Diagnostics

Instrumentation to Chromosomes-HS Structure and disorder-Karyotyping-Chromosomal Studies in Hematological disorders (PBLC and Marrow)FISH - Nucleic acid amplification testing- PCR Principle – types-applications -Thermal cycler-RT PCR-reverse transcriptase PCR-(RT-PCR)-Nested PCR.

Subject XVII
BIO-MEDICAL WASTE MANAGEMENT

Objectives:

To understand the methods and legal of Hospital waste management

Learning Outcome:

1. Will understand the biomedical waste Management
2. To know the procedures of record Keeping of Waste disposal
3. Will understand the health issues in handling biomedical waste
4. To understand the legal aspects of Hospital waste disposal and management

UNIT I **18 Hrs**

Present Scenario

Bio-medical waste – Concepts and Perceptions-Waste Generation-Segregation-Disposal

UNIT II **18 Hrs**

Planning and Objectives of BMW Management-Survey-Policies and Perspectives of BMW Management.

UNIT III **18 Hrs**

Record Keeping-Management of Bio-medical Waste-Technologies for Treatment for BMW-Criteria for selecting appropriate Medical Waste Technologies.

UNIT IV **18 Hrs**

Training-Occupational Safety and Health Issues.

UNIT V **18 Hrs**

Legal Aspects and Environment Concern-Implementation of Action Plan-Approaches to Common Regional facility.